PV102 Day 3: Load Profile and Energy Budget

Students turn a client wish list into a defensible load table, daily watt-hour estimate, priority list, and unknowns register.

PV102 | CTS Day 19 | Instructor Run Sheet
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Core rule

Watts are moments. Watt-hours are work over time.

Learning targets

- Distinguish watts from watt-hours.
- Calculate daily energy use from power and time.
- Rank loads by necessity, runtime, surge risk, and design impact.
- Explain how load decisions affect PV, battery, inverter, and budget.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Calculators Load cards Appliance nameplate examples Student load table Chart paper Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 2 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.

1:00-2:15	Client Load Interview and Energy Budget	Teams interview a mock client, build a daily load table, and recommend one load reduction.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 4 uses load and site information to explore module behavior, strings, and shade impact.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use load profile and energy budget to make the next design decision safer and more defensible. The rule is simple: Watts are moments. Watt-hours are work over time."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Choose one scenario and list all loads.
- For each load, record watts, hours per day, daily Wh, priority, and unknowns.
- Mark surge or motor loads.
- Calculate total daily Wh and essential daily Wh.
- Recommend one load change that improves system practicality without weakening the client need.
- Prepare a handoff showing the load table and top three unknowns.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 4 uses load and site information to explore module behavior, strings, and shade impact.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 3 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 19 | Morning Quiz - Answer Key
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. A 60 W load runs for 3 hours. How much energy is used?

- A. 20 Wh
- B. 60 Wh
- C. 180 Wh
- D. 3 Wh

Answer: C

Rationale/look-for: Energy = watts x hours = 60 x 3 = 180 Wh.

2. Watts measure:

- A. Instant power
- B. Total daily energy only
- C. Wire length
- D. Battery age

Answer: A

Rationale/look-for: Watts are a rate of using or producing energy.

3. Watt-hours measure:

- A. Energy over time
- B. Compass direction
- C. Module frame size
- D. Fuse color

Answer: A

Rationale/look-for: Watt-hours combine power and runtime.

4. The load that often changes inverter selection is:

- A. Sticky note
- B. Surge or motor load
- C. Pencil
- D. Label maker

Answer: B

Rationale/look-for: Starting surge can exceed running watts and affect inverter requirements.

5. A good client load interview asks:

- A. What must run, when, and for how long?
- B. What color is the inverter?
- C. How fast can we buy parts?
- D. Which battery looks best?

Answer: A

Rationale/look-for: Load definition begins with required function and runtime.

6. Calculate: 12 V x 5 A = ___ W.

_____

_____

Answer: 60 W

Rationale/look-for: Power = volts x amps.

7. Why should optional loads be separated from essential loads?

_____

_____

Answer: Priority

Rationale/look-for: Optional loads can be shed to reduce system size, cost, and storage requirements.

8. Name one load unknown that must be verified before final sizing.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept actual watts, surge current, duty cycle, runtime, voltage, season, or user behavior.

9. What happens to required battery capacity when runtime doubles?

_____

_____

Answer: It roughly doubles

Rationale/look-for: If other factors stay constant, more runtime requires more stored energy.

10. Write the Day 3 rule in your own words.

Answer: Power vs energy

Rationale/look-for: Look for distinction between watts and watt-hours.

PV102 Day 3 Whiteboard Outline

PV102 | CTS Day 19 | Whiteboard Outline

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Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Load Profile and Energy Budget Rule: Watts are moments. Watt-hours are work over time. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- Tiny shop: lights, laptop, phone charging, modem/router, small fridge, occasional circular saw.
- Remote trail counter: sensor, camera, cellular modem, winter operation, limited service visits.
- Bus conversion: lighting, fridge, fan, water pump, laptop, phone charging, occasional induction cooktop.

Final board sentence

Watts are moments. Watt-hours are work over time.

PV102 Day 3 Student Packet

Load Profile and Energy Budget

PV102 | CTS Day 19 | Student Print Packet
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Today's rule

Watts are moments. Watt-hours are work over time.

Learning targets

- Distinguish watts from watt-hours.
- Calculate daily energy use from power and time.
- Rank loads by necessity, runtime, surge risk, and design impact.
- Explain how load decisions affect PV, battery, inverter, and budget.

Morning quiz

1. A 60 W load runs for 3 hours. How much energy is used?

- A. 20 Wh
- B. 60 Wh
- C. 180 Wh
- D. 3 Wh

2. Watts measure:

- A. Instant power
- B. Total daily energy only
- C. Wire length
- D. Battery age

3. Watt-hours measure:

- A. Energy over time
- B. Compass direction
- C. Module frame size
- D. Fuse color

4. The load that often changes inverter selection is:

- A. Sticky note
- B. Surge or motor load
- C. Pencil
- D. Label maker

5. A good client load interview asks:

- A. What must run, when, and for how long?
- B. What color is the inverter?
- C. How fast can we buy parts?
- D. Which battery looks best?

6. Calculate: 12 V x 5 A = ___ W.

_____

_____

7. Why should optional loads be separated from essential loads?

_____

_____

8. Name one load unknown that must be verified before final sizing.

_____

_____

9. What happens to required battery capacity when runtime doubles?

_____

_____

10. Write the Day 3 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Client Load Interview and Energy Budget

Teams interview a mock client, build a daily load table, and recommend one load reduction.

Scenario options

- Tiny shop: lights, laptop, phone charging, modem/router, small fridge, occasional circular saw.
- Remote trail counter: sensor, camera, cellular modem, winter operation, limited service visits.
- Bus conversion: lighting, fridge, fan, water pump, laptop, phone charging, occasional induction cooktop.

Team procedure

- Choose one scenario and list all loads.
- For each load, record watts, hours per day, daily Wh, priority, and unknowns.
- Mark surge or motor loads.
- Calculate total daily Wh and essential daily Wh.
- Recommend one load change that improves system practicality without weakening the client need.
- Prepare a handoff showing the load table and top three unknowns.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which load drove the design hardest today: highest watts, longest runtime, or highest priority? Explain.

_____

_____

PV102 Day 3 Afternoon Lab

Client Load Interview and Energy Budget

PV102 | CTS Day 19 | Afternoon Lab
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Mission

Teams interview a mock client, build a daily load table, and recommend one load reduction.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- Tiny shop: lights, laptop, phone charging, modem/router, small fridge, occasional circular saw.
- Remote trail counter: sensor, camera, cellular modem, winter operation, limited service visits.
- Bus conversion: lighting, fridge, fan, water pump, laptop, phone charging, occasional induction cooktop.

Procedure

- Choose one scenario and list all loads.
- For each load, record watts, hours per day, daily Wh, priority, and unknowns.
- Mark surge or motor loads.
- Calculate total daily Wh and essential daily Wh.
- Recommend one load change that improves system practicality without weakening the client need.
- Prepare a handoff showing the load table and top three unknowns.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	
Primary decision	

Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 3 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 19 | Google Classroom Post
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Announcement

Today in PV102 we are working on load profile and energy budget. The core rule is: Watts are moments. Watt-hours are work over time.

Assignment title

PV102 Day 3 - Load Profile and Energy Budget

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.